

Interacting Multipoles of Rank 4 and 4:

[illegible]

Simplify deltas:

[illegible]

[illegible]

[illegible]

Simplify R:

[illegible]

[illegible]

$$\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} - 135135 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\eta\eta} \\
& -270270 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\zeta\zeta} - 405405 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\epsilon\epsilon} \\
& -540540 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{zzz\delta} - 540540 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{zzz\gamma} \\
& -135135 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\gamma\gamma} - 540540 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{zzz\beta} \\
& -270270 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\beta\beta} - 540540 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{zzz\alpha} \\
& -405405 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\alpha\alpha} + 10395 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} + 20790 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\eta\eta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\eta\eta} + 41580 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& +62370 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} + 10395 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\eta\eta} \\
& +124740 \cdot R_z^8 \cdot \Phi_{zz\gamma\delta} \cdot \Phi_{zz\gamma\delta} + 20790 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\zeta\zeta} \\
& +31185 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\epsilon\epsilon} + 20790 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\eta\eta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\eta\eta} + 124740 \cdot R_z^8 \cdot \Phi_{zz\beta\delta} \cdot \Phi_{z\beta\delta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\epsilon\epsilon} \\
& +124740 \cdot R_z^8 \cdot \Phi_{zz\beta\gamma} \cdot \Phi_{zz\beta\gamma} + 41580 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\beta\beta\delta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\beta\beta\gamma} + 41580 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\gamma\gamma} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\eta\eta} + 41580 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} \\
& +62370 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} + 31185 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\eta\eta} \\
& +124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\delta} \cdot \Phi_{zz\alpha\delta} + 62370 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\zeta\zeta} \\
& +93555 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} + 124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\gamma} \cdot \Phi_{zz\alpha\gamma} \\
& +83160 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\alpha\alpha\delta} + 83160 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\alpha\alpha\gamma} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\gamma\gamma} + 124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\
& +83160 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\alpha\alpha\beta} + 83160 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\beta\beta} \\
& +10395 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\alpha\alpha\gamma\gamma} + 20790 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\alpha\alpha\beta\beta} \\
& \quad -945 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\gamma\delta} \cdot \Phi_{\gamma\delta\eta\eta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{zz\gamma\delta} \cdot \Phi_{\gamma\delta\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\gamma\delta} \cdot \Phi_{\gamma\delta\epsilon\epsilon} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\beta\delta} \cdot \Phi_{\beta\delta\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{zz\beta\delta} \cdot \Phi_{\beta\delta\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\beta\delta} \cdot \Phi_{\beta\delta\epsilon\epsilon} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\beta\gamma} \cdot \Phi_{\beta\gamma\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{zz\beta\gamma} \cdot \Phi_{\beta\gamma\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\beta\gamma} \cdot \Phi_{\beta\gamma\epsilon\epsilon} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\zeta\zeta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\eta\eta} - 22680 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\delta} \cdot \Phi_{z\beta\gamma\delta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& \quad -2835 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\alpha\delta} \cdot \Phi_{\alpha\delta\eta\eta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{zz\alpha\delta} \cdot \Phi_{\alpha\delta\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\delta} \cdot \Phi_{\alpha\delta\epsilon\epsilon} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\alpha\gamma} \cdot \Phi_{\alpha\gamma\eta\eta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{zz\alpha\gamma} \cdot \Phi_{\alpha\gamma\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\gamma} \cdot \Phi_{\alpha\gamma\epsilon\epsilon} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\eta\eta} \\
& \quad -7560 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\zeta\zeta}
\end{aligned}$$

$$\stackrel{\text{added up}}{\longrightarrow} R_z^{-17} \cdot \frac{1}{11025} \cdot$$

(3c)

Simplify Hexadecapoles:

$$\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} - 135135 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\eta\eta} \\
& -270270 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\zeta\zeta} - 405405 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\epsilon\epsilon} \\
& -540540 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\delta\delta} - 540540 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\gamma\gamma} \\
& -135135 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\gamma\gamma} - 540540 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\beta\beta} \\
& -270270 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\beta\beta} - 540540 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\alpha\alpha} \\
& -405405 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\alpha\alpha} + 10395 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} + 20790 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{z\delta\eta\eta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{z\delta\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{z\delta\epsilon\epsilon} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{z\gamma\eta\eta} + 41580 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{z\gamma\zeta\zeta} \\
& +62370 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{z\gamma\epsilon\epsilon} + 10395 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\eta\eta} \\
& +124740 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\gamma\gamma} + 20790 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\zeta\zeta} \\
& +31185 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\epsilon\epsilon} + 20790 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{z\beta\eta\eta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{z\beta\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\eta\eta} + 124740 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\beta\beta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\epsilon\epsilon} \\
& +124740 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\beta\beta} + 41580 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{z\beta\beta\delta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{z\beta\beta\gamma} + 41580 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{z\beta\gamma\gamma} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{z\alpha\eta\eta} + 41580 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{z\alpha\zeta\zeta} \\
& +62370 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{z\alpha\epsilon\epsilon} + 31185 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\eta\eta} \\
& +124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} + 62370 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\zeta\zeta} \\
& +93555 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} + 124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\
& +83160 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{z\alpha\alpha\delta} + 83160 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{z\alpha\alpha\gamma} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{z\alpha\gamma\gamma} + 124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\
& +83160 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{z\alpha\alpha\beta} + 83160 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{z\alpha\beta\beta} \\
& +10395 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\alpha\alpha\gamma\gamma} + 20790 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\alpha\alpha\beta\beta} \\
& \quad -945 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\gamma\gamma\eta\eta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\gamma\gamma\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} \\
& -1890 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\beta\beta\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\beta\beta\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\beta\beta\epsilon\epsilon} \\
& -1890 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\beta\beta\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\beta\beta\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\beta\beta\epsilon\epsilon} \\
& -1890 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\zeta\zeta} \\
& -1890 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& -1890 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\eta\eta} - 22680 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\delta} \cdot \Phi_{z\beta\gamma\delta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& \quad -2835 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\alpha\alpha\eta\eta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\alpha\alpha\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\
& 9 \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\alpha\alpha\eta\eta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\alpha\alpha\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon}
\end{aligned}$$

$$\xrightarrow{\text{added up}} R_z^{-17} \cdot \frac{1}{11025} \cdot$$

$$\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} - 135135 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\eta\eta} \\
& -270270 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\zeta\zeta} - 405405 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\epsilon\epsilon} \\
& -540540 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{zzz\delta} - 540540 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{zzz\gamma} \\
& -135135 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\gamma\gamma} - 540540 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{zzz\beta} \\
& -270270 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\beta\beta} - 540540 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{zzz\alpha} \\
& -405405 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zz\alpha\alpha} + 10395 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} + 20790 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\eta\eta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\eta\eta} + 41580 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& +62370 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} + 10395 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\eta\eta} \\
& +124740 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\gamma\gamma} + 20790 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\zeta\zeta} \\
& +31185 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\epsilon\epsilon} + 20790 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\eta\eta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\eta\eta} + 249480 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{z\beta\beta\beta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\epsilon\epsilon} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\beta\beta\delta} + 41580 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\beta\beta\gamma} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\gamma\gamma} + 20790 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\eta\eta} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\
& +31185 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\eta\eta} + 374220 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\
& +62370 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\zeta\zeta} + 93555 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\
& +83160 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\alpha\alpha\delta} + 83160 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\alpha\alpha\gamma} \\
& +41580 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\gamma\gamma} + 83160 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\alpha\alpha\beta} \\
& +83160 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\beta\beta} + 10395 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\alpha\alpha\gamma\gamma} \\
& +20790 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{\alpha\alpha\beta\beta} - 945 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\gamma\gamma\eta\eta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\gamma\gamma\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} \\
& -3780 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\beta\beta\eta\eta} - 7560 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\beta\beta\zeta\zeta} \\
& \quad -11340 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{\beta\beta\epsilon\epsilon} \\
& -1890 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\zeta\zeta} \\
& -1890 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& -1890 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\eta\eta} - 22680 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\delta} \cdot \Phi_{z\beta\gamma\delta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\epsilon\epsilon} \\
& \quad -2835 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\epsilon\epsilon\eta\eta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\epsilon\epsilon\zeta\zeta} \\
& \quad -5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\alpha\alpha\eta\eta} \\
& -11340 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\alpha\alpha\zeta\zeta} \\
& -17010 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\eta\eta} \\
& \quad -7560 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\zeta\zeta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\gamma\eta\eta} \\
& \quad -7560 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
& \quad -1890 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\gamma} \cdot \Phi_{z\alpha\eta\eta} \\
& -22680 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\delta} \cdot \Phi_{z\alpha\gamma\delta} \\
& \quad -3780 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\gamma} \cdot \Phi_{z\alpha\zeta\zeta} \\
& \quad -11340 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\
& \quad -11340 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon}
\end{aligned}$$

(4b)

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot \frac{1}{11025} \cdot$$

$$\begin{aligned} & +2027025 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} - 540540 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{zzz\delta} \\ & - 540540 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{zzz\gamma} - 540540 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{zzz\beta} \\ & - 540540 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{zzz\alpha} + 20790 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\eta\eta} \\ & + 41580 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\ & + 20790 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\eta\eta} + 41580 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ & + 62370 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} + 124740 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\gamma\gamma} \\ & + 20790 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\eta\eta} + 41580 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\zeta\zeta} \\ & + 62370 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} + 249480 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\beta\beta} \\ & + 41580 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\beta\beta\delta} + 41580 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\beta\beta\gamma} \\ & + 41580 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\gamma\gamma} + 20790 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\eta\eta} \\ & + 41580 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} + 62370 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ & + 374220 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} + 83160 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\alpha\alpha\delta} \\ & + 83160 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\alpha\alpha\gamma} + 41580 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ & + 83160 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\alpha\alpha\beta} + 83160 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ & - 1890 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\zeta\zeta} \\ & - 1890 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\eta\eta} - 3780 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ & - 1890 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\eta\eta} - 22680 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\delta} \cdot \Phi_{z\beta\gamma\delta} \\ & \quad - 3780 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\zeta\zeta} \\ & \quad - 5670 \cdot R_z^8 \cdot \Phi_{z\beta\beta\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\ & \quad - 5670 \cdot R_z^8 \cdot \Phi_{z\beta\beta\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\ & \quad - 5670 \cdot R_z^8 \cdot \Phi_{z\beta\gamma\gamma} \cdot \Phi_{z\beta\epsilon\epsilon} \\ & \quad - 3780 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\eta\eta} \\ & \quad - 7560 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\zeta\zeta} \\ & \quad - 3780 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\gamma\eta\eta} \\ & \quad - 7560 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ & \quad - 1890 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\gamma} \cdot \Phi_{z\alpha\eta\eta} \\ & \quad - 22680 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\delta} \cdot \Phi_{z\alpha\gamma\delta} \\ & \quad - 3780 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\gamma} \cdot \Phi_{z\alpha\zeta\zeta} \\ & \quad - 11340 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\ & \quad - 11340 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\ & \quad - 5670 \cdot R_z^8 \cdot \Phi_{z\alpha\gamma\gamma} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ & - 3780 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta} \cdot \Phi_{z\beta\eta\eta} - 7560 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta} \cdot \Phi_{z\beta\zeta\zeta} \\ & - 3780 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta} \cdot \Phi_{z\alpha\eta\eta} - 22680 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\delta} \cdot \Phi_{z\alpha\beta\delta} \\ & \quad - 7560 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta} \cdot \Phi_{z\alpha\zeta\zeta} \\ & \quad - 11340 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ & \quad - 11340 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ & \quad - 22680 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\gamma} \cdot \Phi_{z\alpha\beta\gamma} \\ & \quad + 210 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\gamma\delta} \cdot \Phi_{\gamma\delta\eta\eta} \\ & \quad + 420 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\gamma\delta} \cdot \Phi_{\gamma\delta\zeta\zeta} \\ & \quad + 630 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\gamma\delta} \cdot \Phi_{\gamma\delta\epsilon\epsilon} \\ & \quad + 210 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\delta} \cdot \Phi_{\beta\delta\eta\eta} \\ & \quad + 420 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\delta} \cdot \Phi_{\beta\delta\zeta\zeta} \\ & \quad + 630 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\delta} \cdot \Phi_{\beta\delta\epsilon\epsilon} \\ & \quad + 210 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\gamma} \cdot \Phi_{\beta\gamma\eta\eta} \\ & \quad + 420 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\gamma} \cdot \Phi_{\beta\gamma\zeta\zeta} \\ & \quad + 630 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\gamma} \cdot \Phi_{\beta\gamma\epsilon\epsilon} \\ & \quad + 210 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\delta} \cdot \Phi_{\alpha\delta\eta\eta} \\ & \quad + 420 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\delta} \cdot \Phi_{\alpha\delta\zeta\zeta} \\ & \quad + 210 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\gamma} \cdot \Phi_{\alpha\gamma\eta\eta} \\ & \quad + 420 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\gamma} \cdot \Phi_{\alpha\gamma\zeta\zeta} \\ & \quad + 210 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\gamma} \cdot \Phi_{\alpha\beta\eta\eta} \end{aligned}$$

(5)

Reduce Indices:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot \frac{1}{11025}.$$

[illegible]

(6a)

[illegible]

13

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} - 2162160 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{zzz\alpha} \\
& +582120 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\beta\beta} + 748440 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\
& +415800 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\alpha\alpha\beta} - 45360 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta} \cdot \Phi_{z\alpha\gamma\gamma} \\
& -34020 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\beta\beta\gamma} \\
& -56700 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta} \cdot \Phi_{z\beta\gamma\gamma} \\
& -90720 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\gamma} \cdot \Phi_{z\alpha\beta\gamma} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\gamma\delta} \cdot \Phi_{\beta\beta\gamma\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\delta} \cdot \Phi_{\beta\gamma\gamma\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\gamma} \cdot \Phi_{\beta\gamma\delta\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\delta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\gamma} \cdot \Phi_{\alpha\gamma\delta\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\gamma} \cdot \Phi_{\alpha\beta\delta\delta} \\
& +2520 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\delta} \cdot \Phi_{\alpha\beta\gamma\delta}
\end{aligned} \right] \tag{6c}
\end{aligned}$$

Multiply Out Sum (Duplikate Multipoles):

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot \frac{1}{11025} \cdot \begin{aligned} & +2520 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} + 2520 \cdot R_z^8 \cdot \Phi_{xyyy} \cdot \Phi_{xyyy} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xyyz} \cdot \Phi_{xyyz} + 2520 \cdot R_z^8 \cdot \Phi_{xyzy} \cdot \Phi_{xyzy} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xyzz} \cdot \Phi_{xyzz} + 2520 \cdot R_z^8 \cdot \Phi_{yzxz} \cdot \Phi_{yzxz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{yzyz} \cdot \Phi_{yzyz} + 2520 \cdot R_z^8 \cdot \Phi_{zyyz} \cdot \Phi_{zyyz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xzzz} \cdot \Phi_{xzzz} + 2520 \cdot R_z^8 \cdot \Phi_{xyxz} \cdot \Phi_{xyxz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xyyz} \cdot \Phi_{xyyz} + 2520 \cdot R_z^8 \cdot \Phi_{xyzz} \cdot \Phi_{xyzz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xzzz} \cdot \Phi_{xzzz} + 2520 \cdot R_z^8 \cdot \Phi_{xyzz} \cdot \Phi_{xyzz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} + 2520 \cdot R_z^8 \cdot \Phi_{xxxy} \cdot \Phi_{xxxy} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} + 2520 \cdot R_z^8 \cdot \Phi_{xyxz} \cdot \Phi_{xyxz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} + 2520 \cdot R_z^8 \cdot \Phi_{xyyy} \cdot \Phi_{xyyy} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xyyz} \cdot \Phi_{xyyz} + 2520 \cdot R_z^8 \cdot \Phi_{xyzy} \cdot \Phi_{xyzy} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xyyz} \cdot \Phi_{xyyz} + 2520 \cdot R_z^8 \cdot \Phi_{yzzz} \cdot \Phi_{yzzz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} + 2520 \cdot R_z^8 \cdot \Phi_{xyyy} \cdot \Phi_{xyyy} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xyyz} \cdot \Phi_{xyyz} + 2520 \cdot R_z^8 \cdot \Phi_{xyyy} \cdot \Phi_{xyyy} \\ & +2520 \cdot R_z^8 \cdot \Phi_{yyyy} \cdot \Phi_{yyyy} + 2520 \cdot R_z^8 \cdot \Phi_{yyyz} \cdot \Phi_{yyyz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xyyz} \cdot \Phi_{xyyz} + 2520 \cdot R_z^8 \cdot \Phi_{yyyz} \cdot \Phi_{yyyz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} + 2520 \cdot R_z^8 \cdot \Phi_{xyyz} \cdot \Phi_{xyyz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xyyz} \cdot \Phi_{xyyz} + 2520 \cdot R_z^8 \cdot \Phi_{xyzz} \cdot \Phi_{xyzz} \\ & +2520 \cdot R_z^8 \cdot \Phi_{xyyz} \cdot \Phi_{xyyz} + 2520 \cdot R_z^8 \cdot \Phi_{xyzz} \cdot \Phi_{xyzz} \end{aligned} \quad (7a)$$

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned}
& +2027025 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} - 2162160 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} \\
& +582120 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\beta\beta} + 748440 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} \\
& +748440 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} + 748440 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} \\
& +415800 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\alpha\alpha\beta} - 45360 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta} \cdot \Phi_{z\alpha\gamma\gamma} \\
& -34020 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\beta\beta\gamma} \\
& -56700 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta} \cdot \Phi_{z\beta\gamma\gamma} \\
& -90720 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} - 90720 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} \\
& -90720 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} - 90720 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} \\
& -90720 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} - 90720 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} \\
& -90720 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} + 1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\gamma\delta} \cdot \Phi_{\beta\beta\gamma\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\delta} \cdot \Phi_{\beta\gamma\gamma\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\gamma} \cdot \Phi_{\beta\gamma\delta\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\delta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\gamma} \cdot \Phi_{\alpha\gamma\delta\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\gamma} \cdot \Phi_{\alpha\beta\delta\delta} \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxxx} \cdot \Phi_{xxxx} + 2520 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} + 2520 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} + 2520 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} + 2520 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} + 2520 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} \\
& +2520 \cdot R_z^8 \cdot \Phi_{yyyy} \cdot \Phi_{yyyy} + 2520 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} \\
& +2520 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} + 2520 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} + 2520 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} \\
& +2520 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} + 2520 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} + 2520 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} \\
& +2520 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz}
\end{aligned} \right] \quad (7b)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned}
& +525105 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} + 582120 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\beta\beta} \\
& +491400 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} + 491400 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} \\
& +415800 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\alpha\alpha\beta} - 45360 \cdot R_z^8 \cdot \Phi_{z\alpha\beta\beta} \cdot \Phi_{z\alpha\gamma\gamma} \\
& -34020 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\beta\beta\gamma} \\
& -56700 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\beta} \cdot \Phi_{z\beta\gamma\gamma} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\gamma\delta} \cdot \Phi_{\beta\beta\gamma\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\delta} \cdot \Phi_{\beta\gamma\gamma\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\alpha\beta\gamma} \cdot \Phi_{\beta\gamma\delta\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\delta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\beta\beta\gamma} \cdot \Phi_{\alpha\gamma\delta\delta} \\
& +1260 \cdot R_z^8 \cdot \Phi_{\alpha\beta\gamma\gamma} \cdot \Phi_{\alpha\beta\delta\delta} \\
& +2520 \cdot R_z^8 \cdot \Phi_{xxxx} \cdot \Phi_{xxxx} + 15120 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} \\
& +2520 \cdot R_z^8 \cdot \Phi_{yyyy} \cdot \Phi_{yyyy}
\end{aligned} \right] \quad (7c)
\end{aligned}$$

Multiply Out Sum:

[illegible]

$$\xrightarrow{\text{added up}} R_z^{-17} \cdot \frac{1}{11025} \cdot \left[\begin{aligned} &+1394505 \cdot R_z^8 \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} + 740880 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{zzzz} \\ &+740880 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{zzzz} + 370440 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} \\ &+370440 \cdot R_z^8 \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} - 257040 \cdot R_z^8 \cdot \Phi_{xxzz} \cdot \Phi_{yyzz} \\ &+10080 \cdot R_z^8 \cdot \Phi_{xxxx} \cdot \Phi_{xxxx} + 15120 \cdot R_z^8 \cdot \Phi_{xxxx} \cdot \Phi_{xxyy} \\ &+15120 \cdot R_z^8 \cdot \Phi_{xxxx} \cdot \Phi_{xxzz} + 30240 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} \\ &+15120 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{xxzz} + 15120 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{yyyy} \\ &+15120 \cdot R_z^8 \cdot \Phi_{xxyy} \cdot \Phi_{yyzz} + 10080 \cdot R_z^8 \cdot \Phi_{yyyy} \cdot \Phi_{yyyy} \\ &+15120 \cdot R_z^8 \cdot \Phi_{yyyy} \cdot \Phi_{yyzz} \end{aligned} \right] \quad (8c)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta} \Phi_{\alpha\beta\gamma\delta} \Phi_{\epsilon\zeta\eta\theta} = \left[\begin{aligned} &+ \frac{4427}{35} \cdot R_z^{-9} \cdot \Phi_{zzzz} \cdot \Phi_{zzzz} + \frac{336}{5} \cdot R_z^{-9} \cdot \Phi_{xxzz} \cdot \Phi_{zzzz} \\ &+ \frac{336}{5} \cdot R_z^{-9} \cdot \Phi_{yyzz} \cdot \Phi_{zzzz} + \frac{168}{5} \cdot R_z^{-9} \cdot \Phi_{xxzz} \cdot \Phi_{xxzz} \\ &+ \frac{168}{5} \cdot R_z^{-9} \cdot \Phi_{yyzz} \cdot \Phi_{yyzz} - \frac{816}{35} \cdot R_z^{-9} \cdot \Phi_{xxzz} \cdot \Phi_{yyzz} \\ &+ \frac{32}{35} \cdot R_z^{-9} \cdot \Phi_{xxxx} \cdot \Phi_{xxxx} + \frac{48}{35} \cdot R_z^{-9} \cdot \Phi_{xxxx} \cdot \Phi_{xxyy} \\ &+ \frac{48}{35} \cdot R_z^{-9} \cdot \Phi_{xxxx} \cdot \Phi_{xxzz} + \frac{96}{35} \cdot R_z^{-9} \cdot \Phi_{xxyy} \cdot \Phi_{xxyy} \\ &+ \frac{48}{35} \cdot R_z^{-9} \cdot \Phi_{xxyy} \cdot \Phi_{xxzz} + \frac{48}{35} \cdot R_z^{-9} \cdot \Phi_{xxyy} \cdot \Phi_{yyyy} \\ &+ \frac{48}{35} \cdot R_z^{-9} \cdot \Phi_{xxyy} \cdot \Phi_{yyzz} + \frac{32}{35} \cdot R_z^{-9} \cdot \Phi_{yyyy} \cdot \Phi_{yyyy} \\ &+ \frac{48}{35} \cdot R_z^{-9} \cdot \Phi_{yyyy} \cdot \Phi_{yyzz} \end{aligned} \right] \quad (9)$$